

SNAPDRAGON AI LAB · BUILD & PRESENT CHALLENGE

# VaaniNotes

Offline, multilingual lecture and meeting notes that run on the Snapdragon NPU.

Whisper-Small on the Hexagon NPU

Local LLM via Qualcomm GenieX

11 languages · 0 bytes sent to the cloud

Vaani (वाणी) means speech.

## THE PROBLEM

# Hours of spoken knowledge are lost, or sent to someone else's server

### **Classrooms and meetings are multilingual**

A single lecture moves between Hindi, English and a regional language. Note-taking tools built for English-only audio break down.

### **Connectivity is not guaranteed**

Hostels, trains, rural campuses and field offices have patchy internet. Cloud transcription stops exactly where it is needed most.

### **Audio is private**

Lectures, patient consultations, client calls and internal meetings should not be uploaded to a third party to get a summary.

A Snapdragon X laptop already has the hardware to solve this locally. The NPU just needs the right application.

# Press record. Get notes. Nothing leaves the laptop.

## Transcribes 11 languages

Auto-detects the language, or translates speech straight to English.

## Writes real notes

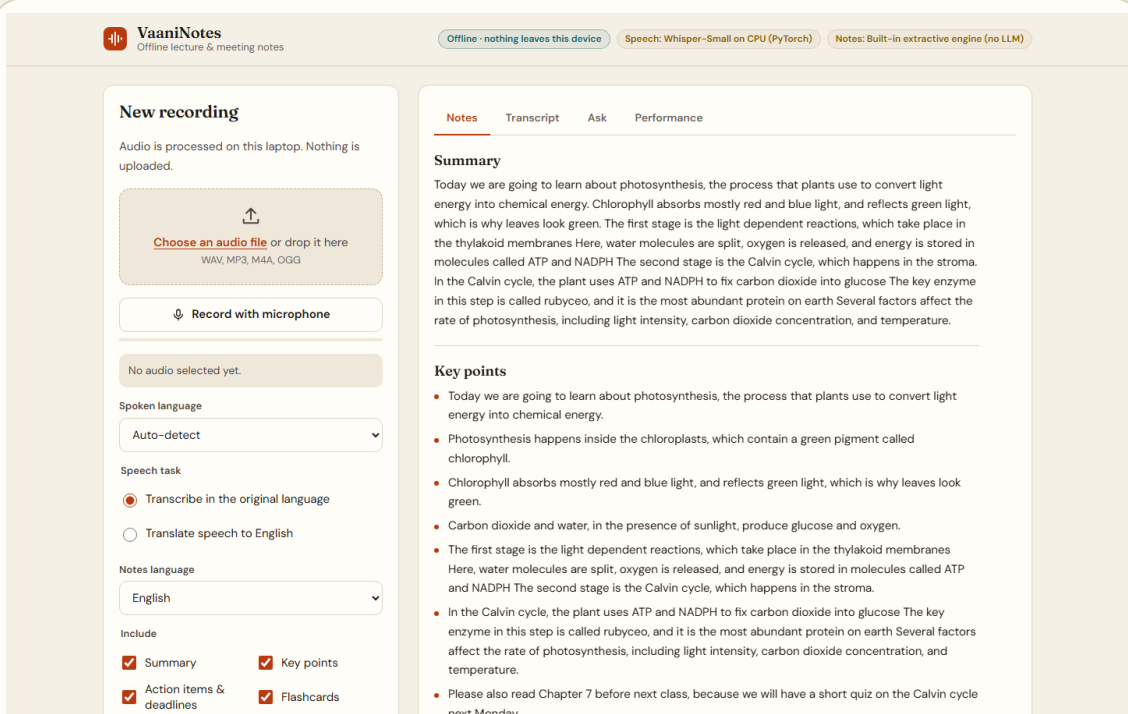
Summary, key points, action items with deadlines, and flip-card flashcards, in the language you choose.

## Answers questions

"When is the lab report due?" returns the answer with its timestamp.

## Private by construction

Binds to 127.0.0.1 with a strict Content-Security-Policy: the page cannot call out.

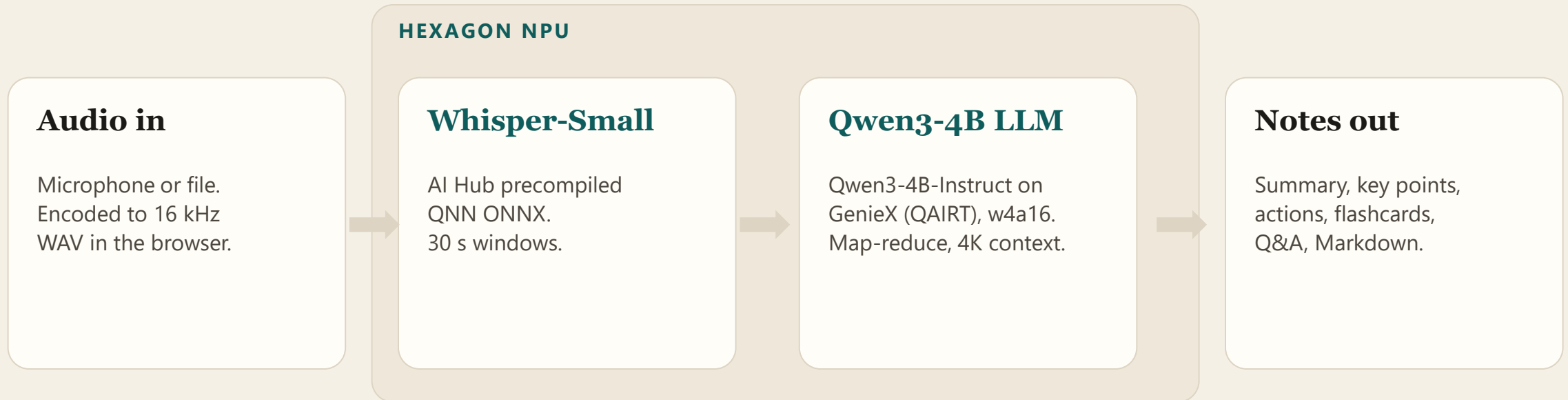


The screenshot displays the VaaniNotes application interface. At the top, it identifies itself as 'VaaniNotes' and 'Offline lecture & meeting notes'. It also shows three status indicators: 'Offline - nothing leaves this device', 'Speech: Whisper - Small on CPU (PyTorch)', and 'Notes: Built-in extractive engine (no LLM)'. The main interface is divided into two panels. The left panel, titled 'New recording', contains instructions that audio is processed on the laptop and is not uploaded. It offers options to 'Choose an audio file or drop it here' (supporting WAV, MP3, M4A, OGG) or to 'Record with microphone'. Below this, it states 'No audio selected yet.' and provides settings for 'Spoken language' (set to 'Auto-detect'), 'Speech task' (with options for 'Transcribe in the original language' and 'Translate speech to English'), and 'Notes language' (set to 'English'). At the bottom of this panel, there are checkboxes for 'Include' options: Summary, Action items & deadlines, Key points, and Flashcards, all of which are checked. The right panel shows the generated notes under the 'Notes' tab. It includes a 'Summary' section with a paragraph about photosynthesis, a 'Key points' section with a bulleted list of key concepts, and a 'Performance' section with a note about a short quiz on the Calvin cycle.

The app turning a 93-second biology lecture into notes (development build, CPU fallback).

## HOW IT WORKS

# One pipeline, two models, all on the NPU



FastAPI on 127.0.0.1 runs a single-worker job queue and streams results to the page as they are produced: transcript first, then each notes section.

Fallback chain keeps it useful anywhere: NPU → CPU for speech; GenieX → any local OpenAI-compatible server → built-in extractive engine for notes.

# Optimised for Snapdragon, not just ported to it

## Chipset-specific AI Hub models

setup.ps1 detects X Elite, X Plus or X2 Elite and fetches the matching precompiled QAIRT context binaries for Whisper-Small from Qualcomm AI Hub.

## ONNX Runtime + QNN on HTP

Encoder and decoder run through the QNN execution provider in burst mode. Both the plugin EP (2.x) and the legacy built-in EP are supported.

## A PyTorch-free NPU path

The KV-cache decoding loop is written in numpy against the AI Hub I/O contract (float16, named tensors), so the Snapdragon install stays small and native ARM64.

## LLM on the NPU via GenieX

Qwen3-4B-Instruct (w4a16, QAIRT) is reached over GenieX's local OpenAI-compatible endpoint. Long recordings are condensed map-reduce style to fit 4K tokens.

## PERFORMANCE

# About 30× faster than real time on the NPU

**117 ms**

Whisper-Small encoder  
per 30 s of audio

**10.5 ms**

Whisper-Small decoder  
per token

**22.6 tok/s**

Qwen3-4B-Instruct  
GenieX QAIRT, w4a16

Published by Qualcomm AI Hub for Snapdragon X Elite (NPU), models v0.63.0. On X2 Elite: 53 ms / 6.3 ms.

30 s of speech  $\approx 117 \text{ ms} + 80 \text{ tokens} \times 10.5 \text{ ms} \approx 1 \text{ s}$  of NPU time.

A one-hour lecture transcribes in about two minutes, then notes are written at roughly 20 tokens per second.

Measured CPU fallback on a 13th-gen Intel i5: 1.7–2.1× real time for the same model.

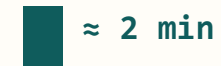
### Time to transcribe a 60-minute lecture

CPU fallback (measured, i5)



$\approx 30 \text{ min}$

Snapdragon X Elite NPU (from AI Hub figures)



$\approx 2 \text{ min}$

# Two commands to install. Usable by everyone.

## Deployment

```
powershell -File setup.ps1  
.\run.ps1
```

- Detects the chipset and downloads the right AI Hub model
- No API keys, no accounts, no cloud services
- Works in airplane mode after setup; fonts are self-hosted
- CPU and no-LLM fallbacks, so it runs on any Windows PC
- Web UI, command line, and Markdown export

## Accessibility

- Full keyboard operation, visible focus, skip link
- Screen-reader landmarks, ARIA tabs, live progress updates
- WCAG AA contrast, 44 px targets, 200% zoom safe
- Light and dark themes; respects reduced motion
- Notes in 11 languages with correct lang tags, so Indic scripts get the right font and voice
- Status is never colour-only: icon + words every time

# Who gets their hours back

## Students

Lectures in mixed Hindi-English become revision notes and flashcards, even in a hostel with no Wi-Fi.

## Teachers

Class recordings turn into summaries and homework lists to share with absentees.

## Clinics & legal aid

Consultations stay confidential because audio never leaves the device.

## Field teams

Government and NGO staff get minutes and action items in low-connectivity offices.

## What makes it different

Indian languages first, not an afterthought · Both models on the NPU, not just one · Provably offline (CSP + loopback) · Degrades gracefully instead of failing

# What is built, what is verified, what is next

## Built

- NPU speech path (QNN EP)
- GenieX LLM client + fallbacks
- Notes, flashcards, Q&A
- Web UI, CLI, Markdown export
- One-step setup, benchmark script

## Verified

- 11 unit tests, including the AI Hub decoder I/O contract
- End-to-end pipeline, API and UI on the CPU path
- Model I/O checked against the published AI Hub assets
- Still to do on a Snapdragon laptop: NPU timings with scripts/benchmark.py

## Next

- Whisper-Large-v3-Turbo from AI Hub for higher Indic accuracy
- Live captions while recording
- Speaker diarisation
- Packaged installer (MSIX)

# Your lecture stays yours.

VaaniNotes turns the Snapdragon NPU into a private note-taker for India's multilingual classrooms and workplaces.

Source code, setup guide, design system and benchmarks are in the GitHub repository linked with this submission.

Models: Whisper-Small and Qwen3-4B-Instruct via Qualcomm AI Hub. Snapdragon and Qualcomm are trademarks of Qualcomm Incorporated.